

NUTS & BOLTS — COMPLETE LIST

REV 011b

2026-08-24

Apollo Track Pod · two pods, one per side · nothing below class 8.8 anywhere · grade 8 (US) = class 10.9

source: FASTENERS.md +
apollo_track_pod_rev011.scad

A · PIVOT AXLE

Sheet 2 · §7.2 / §7.3

ITEM	/POD	BOTH
M16 × 195 hex bolt , part-threaded 45 mm one end, cl.10.9 (the pivot axle both arms swing on; smooth shank 150 must span all 4 bushings + spacer — bushings never ride on thread; head stamped 10.9)	1	2
M16 nylock nut (closes the pivot stack — one per bolt only, it is a headed bolt; zero end-float then back 1/8 turn, no torque figure)	1	2
Ø16 hardened flat washer (under the pivot bolt head and under the nylock)	2	4
Ø16 × Ø30 × 1.5 hardened thrust washer (between every steel face that rotates against another in the pivot stack: spacer-flange, fork-fork scissor gap, flange-sleeve)	6	12

Substitute if the bolt can't be found: M16 double-end stud × 190 + 2 nylocks per pod, smooth middle ≥ 150. Prefer 8.8/B7 over stainless. Companions in this stack (not fasteners): 4 flanged bronze bushings SAE 841 Ø16×Ø22×20 per pod.

B · IDLER WHEEL AXLES

supersedes §7.5 · studs purchased 2026-07-25

ITEM	/POD	BOTH
GB901 M12 × 110 double-end stud , 8.8 (TRAILING idler axle — runs in the 25 mm tensioner slot)	1	2
GB901 M12 × 120 double-end stud , 8.8 (LEADING idler axle — plain Ø15.4 bore; longer because the leading fork is wider)	1	2
M12 nylock nut (two per stud — a double-end stud has no head)	4	8
M12 washer , ≈2.5 thick, snug Ø13 bore (caps the Ø15 sleeve column AND clamps the plate — goes directly against the plate, ALWAYS before the eye nut; a sloppy bore that slips over the Ø15 ring lets the fork crush onto the wheel)	4	8

Sleeves (inner tube 50.8 TR / 66.5 LD + 2 × 6.35 rings per axle) come from the 500 mm Ø15×Ø12 pipe in the §7.1 steel order — cut long, file square, fit at dry-stack. Pass test: wheel spins free with zero side-play at full torque; if it drags, the tube is short — never back the nut off instead.

C · FORK MOUNT

§7.7

ITEM	/POD	BOTH
M8 × 30 bolt , cl.10.9 (bolts each carrier to its fork leg: carrier 6 + leg 4 + nut; locks the torque-arm rotation)	2	4
M8 nylock nut (on the fork-leg bolts)	2	4
M10 hub-motor axle nut (holds the hub motor in the carrier's flatted Ø10.4 torque-arm hole — REUSED from the donor pod, do not buy)	—	reuse

D · SHOCKS

§7.7 / §9.5 · 4 shocks total, eyes measured Ø8

ITEM	/POD	BOTH
M8 part-threaded bolt , ≈135 (lower shock eye — through both arm plates at station a; the eye rides the spacer sleeve on the smooth shank)	2	4
M8 nylock nut (on the lower shock bolts)	2	4
M8 pin/bolt ≈35–40 + nylock (UPPER shock eye, through the Ø8.4 hole in the carrier tab — missing from §7.7; length unspecified anywhere, measure at the §9.5 fitting with the real shock eye in hand)	2	4

Companion (steel order, not a fastener): Ø15 OD × Ø9 ID × ≈45 spacer sleeve for each lower eye — 4 total. Confirm the ≈135 lower length at §9.5 too.

E · BELT TENSIONER

Rev 011b PUSHER · supersedes §7.6 pull-type · trailing arm only

ITEM	/POD	BOTH
M6 × 45 full-thread push bolt , cl.8.8+ (cut the purchased ×60 down to 45; threads through the block welded FORWARD of the slot — screwing in PUSHES the axle rearward, loaded in compression; advance both sides evenly. Rev 011b: replaces the pull-type whose head hit the belt teeth)	2	4
M6 jam nut (locks the push bolt against the block's forward face once belt tension is set)	2	4
M6 plain nut, welded to the block (the thread the push bolt runs in — weld over the Ø6.6 hole on the forward face (or drill 5.0 and tap M6 instead))	2	4
M6 lifting eye nut, DIN 582 (NO LONGER NEEDED — replaced by the Ø15 × 7 push collar cut from the axle-sleeve pipe; keep the owner's eye nuts as spares)	—	—

F · THREADED CONSUMABLES

§7.8

ITEM	/POD	BOTH
M6 grease zerk (one per fork boss if you add the recommended cross-drilled grease passage — verify the thread really is M6)	1	2
Nylock + hardened washer assortment , M8/M10, class 10 / Gr8 (for the fitting-up you will inevitably redo)		1 box
Blue (medium) threadlocker + paint pen (threadlocker on non-nylock threads; pen for torque witness marks on every fastener)		1 ea

DO NOT BUY — SUPERSEDED BY REV 011

M8 threaded rod ~172 + 4 nuts + washers (\$7.6 keel standoff)	keel is deleted — fork legs box the carriers
M8 keel bolt through the carriers	Rev 011 carrier strips are 220 with no keel hole
4 × M15×1.5 fine nuts + 8 Ø15 washers (\$7.5)	idler axles are now M12 studs; Ø15 shaft kept only as spare
Second M16 nylock per pivot bolt	the bolt has a head — one nut per bolt
Anything below class 8.8	unmarked hardware-store bolts are out, everywhere

TORQUE

M6 (draw bolts / jam nuts)	snug + jam nut — sets belt tension, not preload	
M8 cl.10.9	30 N·m	
M10 (hub axle, reused nuts)	60 N·m	
M12 (idler axle nylocks)	105 N·m	— only after the wheel spins free with zero side-play
M16 (pivot nylock)	zero end-float, then back off ⅛ turn — running fit, never a torque figure	

Paint-pen witness marks on everything. Retorque at 1 h / 5 h / then every 20 h (\$10).

SHOPPING CHECKLIST — BOTH PODS

☐ M16 × 195 part-thread cl.10.9 bolt	2	☐ M8 part-thread bolt ≈135	4
☐ M16 nylock nut	2	☐ M8 pin/bolt ≈35–40 (measure at \$9.5 first)	4
☐ Ø16 hardened flat washer	4	☐ M8 nylock nut	12
☐ Ø16×Ø30×1.5 thrust washer	12	☐ M6 × 60 full-thread bolt 8.8 (cut to 45 — push bolts)	4
☐ M12 × 110 stud 8.8 (have — purchased)	2	☐ M6 jam nut	4
☐ M12 × 120 stud 8.8 (have — purchased)	2	☐ M6 plain nut (weld to pusher blocks)	4
☐ M12 nylock nut	8	☐ M6 eye nut DIN 582 (Rev 011b: not needed — spares)	0
☐ M12 washer, snug Ø13 bore	8	☐ M6 grease zerk	2
☐ M8 × 30 cl.10.9 bolt	4	☐ M10 hub axle nut (reuse from donor)	—

Assembly order that matters: on the idler axles the M12 washer goes directly against the arm plate, then the Ø15 push collar (trailing only, Rev 011b), then the nylock — washer first is what caps the Ø15 sleeve column; a part with a bore bigger than Ø15 placed against the plate leaves the column uncapped and the fork can be crushed onto the wheel.